Early Diagnosis of Brain Tumor from MRI Reports Using LLMs with RAG

Abstract and Diagnostic Workflow

Early detection of intracranial lesions is paramount for optimizing therapeutic efficacy and patient survival outcomes. Magnetic Resonance Imaging (MRI) serves as the primary non-invasive diagnostic modality. Standard imaging sequences include T1-weighted, T2-weighted, Fluid-Attenuated Inversion Recovery (FLAIR), and T1 contrast-enhanced (T1CE) imaging using gadolinium.

MRI Key Diagnostic Signatures

Glioma MRI Features: Typically display heterogeneous hypointensity on T1, hyperintensity on T2/FLAIR, and irregular peripheral ring enhancement on T1CE with extensive surrounding vasogenic edema and mass effect.
Meningioma MRI Features: Well-circumscribed extra-axial dural-based mass, intense uniform contrast enhancement on T1CE, presence of a CSF cleft sign separating the mass from brain parenchyma, and a classic dural tail sign.
Pituitary Adenoma MRI Features: Sellar or suprasellar mass causing expansion of the sella turcica, variable T1/T2 signal, and homogenous or heterogeneous enhancement with optic chiasm elevation.
Normal Brain MRI Features: Intact symmetrical cerebral hemispheres, normal ventricles and sulci, sharp grey-white matter differentiation, no abnormal contrast enhancement or mass displacement.

LLM & RAG Integration in Clinical Decision Support

Integrating Convolutional Neural Network (CNN) image classification models with Retrieval-Augmented Generation (RAG) language assistants enables a dual-path pipeline: high-precision automated MRI slice classification combined with instant context-grounded medical explanation. RAG retrieves relevant medical literature to provide empathetic, evidence-based answers to patient questions, mitigating anxiety while ensuring strict adherence to verified medical facts.